

Yongcan Huang, PhD

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Data Scientist | Machine Learning & Deep Learning | Foundation Models & LLMs | Time-Series Forecasting

PROFESSIONAL SUMMARY

Data scientist (Ph.D.) with 7 years of combined research and industry experience building, validating, and communicating machine learning, deep learning, and foundation-model solutions on large, multi-source data. Lead author of a benchmark of six time-series foundation model configurations (TimesFM, Chronos-2, Moirai-2, Time-MoE, including LoRA fine-tuning) against trained deep learning models on 1.7M records, and author of peer-reviewed work on graph autoencoders, contrastive learning, and LLM-based classification. Experienced across the full model lifecycle: data acquisition and pipeline engineering, leakage-aware validation design, evaluation on highly imbalanced targets, and translating results into decisions for non-technical stakeholders. Program committee member for WSDM, PAKDD, ACML, and WISE.

EDUCATION

Ph.D. in Engineering, Resilient Infrastructure Systems

Dec 2024

University of Georgia, Athens, GA • Selected as College of Engineering Convocation Speaker

M.S. in Traffic and Transportation Engineering

Dec 2019

Chanhsha University of Science and Technology, Changsha, China

B.S. in Civil Engineering

Jul 2017

Wuhan Institute of Technology, Wuhan, China

MACHINE LEARNING RESEARCH

UMBC – Big Data Analytics Lab (Voluntary Visiting Scholar) – Baltimore, MD

2026 – Present

Lead Author | Benchmarking Time-Series Foundation Models for Wildfire PM2.5 Forecasting (arXiv:2607.07951)

- Benchmarked six foundation model configurations (zero-shot TimesFM, Chronos-2, Moirai-2, Time-MoE; LoRA fine-tuned Chronos-2 and Time-MoE) against trained LSTM, BiLSTM, and Transformer models and a persistence baseline for 6-, 12-, and 24-hour forecasting, running the foundation model experiments on AWS
- Built a 12-year (2013–2025) hourly benchmark with an asynchronous, incident-aware crawler that paired 1,375 wildfire incidents to the nearest of 79 EPA monitors, producing 1.73M records and 277,866 training windows; segmented series at sensor gaps instead of interpolating to avoid injecting artificial dynamics
- Designed a severity-stratified leave-one-incident-out cross-validation protocol that keeps each fire event entirely in one fold, removing the within-event leakage that chronological splits introduce
- Fine-tuned foundation models with LoRA adapters trained per fold on training incidents only, fixing a zero-shot Chronos-2 tail instability (R^2 from -0.08 to 0.60 ; RMSE down $7.8 \mu\text{g}/\text{m}^3$)
- Evaluated with MAE, RMSE, R^2 , and exceedance F1 at EPA AQI thresholds on a heavily imbalanced target (0.28% hazardous hours); showed a compact BiLSTM beat every foundation model (hazardous F1 0.63 vs. ≤ 0.54) and translated results into guidance on when to fine-tune versus train a specialist model

University of Georgia – Smart Mobility & Infrastructure Lab – Athens, GA

Jan 2021 – Dec 2024

Graph Deep Learning for Traffic Data Imputation and Fault Detection

- Developed a cluster-guided denoising graph autoencoder that groups traffic sensors by dual-encoded spatial-temporal correlation and reconstructs missing or faulty readings, improving imputation and fault detection (Expert Systems with Applications, 2025)
- Engineered a symmetric contrastive learning triplet network with efficient contrastive sampling and a cross-attention-boosted loss for fault detection in GDOT sensor time series, outperforming benchmark methods (International Journal of Data Science and Analytics, 2024)
- Deployed a Streamlit application for real-time traffic data monitoring and automated fault detection

Crash Severity Classification with Large Language Models

- Applied GPT-3.5-turbo and LLaMA-3 to multi-class crash severity classification by converting tabular crash records into text narratives, using chain-of-thought prompting and prompt engineering to improve accuracy and interpretability (Computers, 2024)

Clemson University – NSF-funded Research – Clemson, SC

Jan 2020 – Dec 2021

- Built and tested a convolutional neural network that recognizes construction-site collision hazards from labeled audio data

PROFESSIONAL EXPERIENCE

Brudis & Associates (Traffic Engineer, Data Science & Modeling) – Columbia, MD *Jun 2025 – Present*

Micro-Transit Acceptance Modeling | Baltimore County, MD

- Designed a resident survey (2,400+ responses) from a systematic review of demand factors and structured it into 41 modeling features (multi-select and ordinal items); joined Census, land use, and transit service data in ArcGIS Pro to locate transportation-dependent populations across seven rural communities
- Framed resident acceptance of micro-transit as a binary classification problem on an imbalanced target (8.5% positive interest), using a stratified, duplicate-aware train/test split to prevent leakage between identical responses
- Trained logistic regression and XGBoost classifiers; logistic regression reached PR-AUC 0.77 and ROC-AUC 0.97, with predicted acceptance within 0.5 percentage points of the observed rate
- Benchmarked these models against five open-weight LLMs (Qwen, Llama, Mistral, Gemma, Phi) under zero- and few-shot prompting, scoring LLMs by next-token Yes/No probabilities; the best LLM (Llama-3.1-8B, 9-shot) reached PR-AUC 0.86 (mean of 3 exemplar seeds) vs. 0.77 for logistic regression

Traffic Operations & Safety Analysis

- Produced traffic demand forecasts, Synchro/VISSIM capacity models, and ISATe/HSM crash prediction inputs for multiple project alternatives analysis submitted to MDOT SHA and FHWA

Stantec Consulting Services Inc. (Traffic Engineer) – Lake Mary, FL *Jan 2025 – May 2025*

- Cleaned and spatially joined crash records to roadway profiles and ran high-injury network analysis to identify crash hotspots, feeding Safe System–based safety action plans and road safety audit recommendations for public agency clients
- Ran corridor operational (Synchro, HCM) and travel demand model (Cube Voyager) scenario analyses

Arcadis (Transportation Analyst Intern) – Atlanta, GA *May 2024 – Aug 2024*

- Built Python and ArcGIS Pro dashboards comparing corridor-level performance across build and no-build alternatives
- Built a Python/Streamlit web application that automates Synchro output transformation and visualization, reducing manual processing time for engineering teams

TECHNICAL SKILLS

- **Programming & Data:** Python, R, SQL (MySQL), MATLAB, Git/GitHub
- **Cloud & Big Data:** AWS, Apache Spark, H2O
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LoRA fine-tuning
- **Foundation Models & LLMs:** TimesFM, Chronos-2, Moirai-2, Time-MoE, GPT, LLaMA, prompt engineering, chain-of-thought
- **Methods:** classification, clustering, time-series forecasting, graph neural networks, autoencoders, contrastive learning, cross-validation design
- **Visualization & Geospatial:** Tableau, Power BI, Streamlit, ArcGIS Pro, QGIS

SELECTED PUBLICATIONS

1. **Huang, Yongcan**, Li Jiang, and Ze Yu Liu. "Evaluating the Generalizability of Foundation Models for Extreme Environmental Events: Case Study of California Wildfire PM2.5." *arXiv preprint arXiv:2607.07951* (2026).
2. **Huang, Yongcan**, Hao Zhen, and Jidong J. Yang. "Cluster-guided denoising graph auto-encoder for enhanced traffic data imputation and fault detection." *Expert Systems with Applications* 261 (2025): 125531.
3. Zhen, Hao, Yucheng Shi, **Yongcan Huang**, Jidong J. Yang, and Ninghao Liu. "Leveraging Large Language Models with Chain-of-Thought and Prompt Engineering for Traffic Crash Severity Analysis and Inference." *Computers* 13, no. 9 (2024): 232.
4. **Huang, Yongcan**, and Jidong J. Yang. "Symmetric contrastive learning for robust fault detection in time-series traffic sensor data." *International Journal of Data Science and Analytics* (2024): 1-15.

Full publication list available upon request.

HONORS & SERVICE

- Lifesavers National Conference Traffic Safety Scholar (2023, 2025); APHA ICEHS President's Road Safety Scholarship (2023)
- Program Committee Member: WSDM 2027, PAKDD 2025 & 2026, ACML 2026, WISE 2026; Peer Reviewer, TRBAM-27; Corresponding Member, ASCE AI in Transportation and Data Sensing & Analytics committees